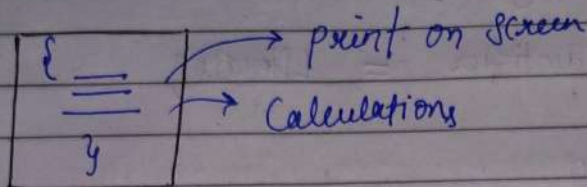


DAY-03

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Variable and Datatype;

① Variable



Demo.java

$$\frac{X}{2} + \frac{Y}{3} = ?$$
$$2 + 3 = 5.$$

x	y		
2	3		

Memory.

FirstNum → 2
Container

SecondNum → 3
Container

↑
Identifiers

→ Think as the code of Black Box.

class in java →

```
public class Demo {  
    public static void main(String[] args) {  
        }  
    }  
}
```

Data Type

→ Java is statically typed.

Declaration

Data Type Identifier = Literal;

Data Type

Primitive

Non-primitive (forward →)

- Integer → byte, short, int, long
- Real No → float, double
- Character → char
- Boolean → boolean

① Integer



Final Num = 10;

byte short int long

(on basis of storage
different Data type
in integer occurred)

(-128 to 127)

range

③ Binary Representation

$(4)_{10}$

$(0-9) \rightarrow 10, 11, 12, 13 \dots$

Decimal Number

$0-1 \rightarrow 0, 1$

Binary Number

Name	width	Range
long	64	-9,223,372,036,854,777,808 to 9,223,372,036,854,777,807
int	32	-2,147,483,648 to 2,147,483,647
short	16	-32,768 to 32,767
byte	8	-128 to 127

In Java

Every Number is signed

Combinations

00 - 0
10 - 2
11 - 3

$2^1 - 1 = 1$

$2^2 - 1 = 3$

②

for $n=3$

$2^3 - 1 \Rightarrow 7$ is biggest No.

for byte.

① →

1	2	3	4	5	6	7	8
1	2	3	4	5	6	7	8

② # how to represent -ve Number in next day.

② Real Number.

eg.

5.23
10.02

Floating point Number

Float double

Name width in bits

Range

double

64

4.9e-324 to 1.8e+308

float

32

1.4e-045 to 3.4e+38

float - single precision

double - double precision.

③ Character

Char → unicode character.

→ ASCII → it provides digit to special character

a-z

A-Z

0-9

-, , " , ' , (,)

→ alphabet →

a

container

* use single quotes to store it in { alphabet variable }
char alphabet = 'a';

'a' → 65 →

65

character

Java → unicode

* (it provides integer value for all world wide characters)

Char → 16 bit

④ Boolean.

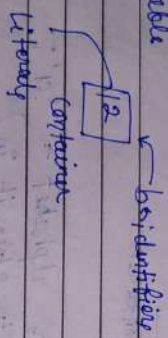
boolean

true false

Syntax

boolean b = false;

variable



Have understand Binary, Octal, Hexadecimal.

→ byte b = 0b101;

byte b1 = 05;

byte b2 = 0x1A8;

Access specifier

int x = 5;

declaring & defining at the same time.

Keywords

System.out.println();
class
static
void
public
→ "Keywords are pre reserved word for Java"
or
"Total 68 keywords in Java"

int x; ← declaration

x = 9 ← definition

